DisasterM3 vs BRIGHT — Optical Data Audit & EDA

Easy English version | Report date: Oct 19, 2026 | @Ridita Afrin

What is this? This is the same report, written in very easy English. The numbers are the same as in the original PDF.

How to read it: The **bold words** are the words used in the original PDF. The grey tag next to each heading (for example original p. 4) shows the page in the original PDF.

Two parts: Part 1 is the main report. Part 2 lists the mistakes in an older PDF, with the right numbers.

Hard words first

Word	Easy meaning	Banglish
Dataset	A big set of photos and labels	data er songroho
Optical image	A normal color photo from a satellite	rongin chobi
SAR image	A radar picture. It is black and white.	radar chobi
Pre image	A photo taken **before** the disaster	durjoger ager chobi
Post image	A photo taken **after** the disaster	durjoger porer chobi
Pair	One pre photo + one post photo of the same place	jora
Tile	One square photo, 1024 × 1024 pixels	chobir tukra
Label / mask	A picture that marks each pixel: building or not, broken or not	
Intact / damaged / destroyed	not broken / a little broken / fully broken	thik / ektu bhanga / puro bhanga
Train set	The photos the model **learns** from	shekhar data
Val set / frozen val	The photos we use to **test** the model. "Frozen" means we never change it.	porikkhar data
Leak	Some test photos are also in the learning photos. So the model saw the answers before the test.	proshno faash
Pool / optical pool	The 6,443 color pairs that our champion model uses	
Audit	A careful check to find problems	
EDA	Studying data with numbers and charts	

Word	Easy meaning	Banglish
Tag	A small note saved with a file, for example "this is SAR"	
Resolution / GSD / m per pixel	How much ground one pixel shows. 0.3 m means one pixel shows 30 cm. A smaller number means more detail.	
Byte for byte / byte-identical	Two files are exactly the same	hubohu same
Median	The middle value, after you sort all the values	majher value
IoU	A score from 0 to 1. It shows how well two masks cover the same area. 1 means a perfect match.	
Destroyed IoU	How well the model finds destroyed buildings	
Gate	The score we must pass. For us: Destroyed IoU above 0.30	target

Part 1 · The main report

Title and first paragraph

original p. 1

"DisasterM3 vs BRIGHT — Optical Data Audit & EDA": We checked two datasets, DisasterM3 and BRIGHT. We looked only at color photos. We looked for problems, and we studied the data.

"Oct 19, 2026 · @Ridita Afrin": This is the date and the writer.

First paragraph:

- We can use both datasets for our color-photo model.
- But we found 2 big things:
  1. DisasterM3 has a **leak** in its test set.
  2. Our 5 BRIGHT events can teach the model **where the buildings are**. They **cannot** teach it what destroyed buildings look like.

The 7 main points

original p. 1

Each point is explained again later.

1. **"Validation leak."** There are 645 test pairs. 40 of them (6.2%) are almost copies of learning pairs. This is because two xBD events, `socal_fire` and `woolsey_fire`, have the same 264 pre photos. These 40 pairs hold 8.6% of all destroyed pixels in the test set.
2. **"Our BRIGHT subset does not leak."** Our 5 events have 1,285 tiles. **None** of them is in the optical pool. 1,165 of them are in DisasterM3, but only as SAR pairs or Bench masks. So they are **safe**.
3. **"BRIGHT has no optical post image."**
   - All 4,246 BRIGHT post photos are SAR, so we cannot use them.
   - Our 5 events have 21% destroyed labels, but we cannot use those labels.
   - We get only the **pre** photos. Before the disaster, every building was fine. So the buildings go in as "intact".
   - 82% of these tiles are volcano photos. But the test set has only 8 volcano pairs.

4. **"The papers' resolutions are wrong."**
    - The BRIGHT files say one resolution. The BRIGHT paper says another. They do not match for 6 of the 14 events.
    - The DisasterM3 paper says 0.8 m. But its 2,828 copies of BRIGHT tiles are 0.30 m.
5. **"Appearance gap."** Our 5 events look different from the other photos. They are **darker** (middle brightness 60, but 84 in the pool), and they are **2.6 times sharper**.
6. **"Pairs are not 'strictly aligned'."** The paper says the pre and post photos match exactly. They do not. In 52% of the pairs we could measure, one photo is moved by **more than 2 pixels**, and in some pairs by up to **56 pixels**.
7. **"SAR hides behind file names."** In 73% of SAR post files, the name has no "sar". The tags are right and the pixels are right. Only the names are wrong.

"Scope and SAR exclusion": what we checked, and how we removed SAR

original p. 1–2

- We counted every number ourselves, on the data on our computer. We did not copy numbers from the papers.
- We removed SAR by **looking at the pixels**, not only by reading the tags.

DisasterM3 has 2 parts: **Instruct** (photos for learning) and **Bench** (photos for testing, a "benchmark").

Source	What we used	Size
DisasterM3 Instruct images	Pre photos and color post photos	20,988 files
DisasterM3 Instruct masks	Building and road masks of the pool	6,443
DisasterM3 Bench	Only its list file (JSON). **The photos are not on our computer.**	30,042 entries
BRIGHT pre-event	Color tiles and their file tags	3,395 tiles, 11 events
BRIGHT post-event	**Not used.** All of them are SAR.	4,246
BRIGHT targets	Label maps: 0 = no building, 1 = intact, 2 = damaged, 3 = destroyed	4,246

How we find SAR, pairs, and the pool

original p. 2

"SAR rule." How do we know if a photo is SAR?

- A color photo has 3 channels: red, green and blue. In a SAR photo, the 3 channels are the same.
- So if the 3 channels are almost the same (the difference is less than 1 grey level), the photo is SAR.
- We check only the pixels that have data.
- We also compare our result with the DisasterM3 tags (`image_type` / `post_image_type`).
- We need the pixel test because **2,066 files have no tag**.

"Pairs." The pairs come from the file `train_release.json`. It has **7,160 color pairs** and **2,648 SAR pairs**, and they use 9,117 pre photos.

"Optical pool." Our champion model uses **6,443 color pairs**. They are split with `random_state=42` into **5,798 for learning** and **645 for testing**. The 645 is the "frozen val" of our E-protocol. We never change it.

"BRIGHT's optical side is pre-event only."

- BRIGHT has color photos only from **before** the disaster.
- The folder name is `pre-event_wo_ukraine_myanmar_mexico`. Here "wo" means "without".
- So the photos of Ukraine (513), Myanmar (126) and Mexico (212) are **not in this folder**. These 3 events have labels and SAR, but no color photo here.

"Inventory": how many photos of each kind

original p. 2–3

- Using the pixel test: DisasterM3 Instruct on our computer has **17,451 color photos** and **3,537 SAR photos**.
- BRIGHT adds **3,395 color pre tiles**. BRIGHT adds **no** color post photo.

Part	Pre (color)	Post (color)	Post (SAR)	Notes
DisasterM3 Instruct, all files on disk	9,746	7,705	3,537	Total 20,988. The paper's total is the same. But the paper says 17,190 color and 3,798 SAR, and our count is different.
DisasterM3 Instruct, files named in the JSON list	9,116	7,159	2,647	**2,066 files are not in the list**
Optical pool (our model)	6,443	6,443	0	5,798 learning, 645 test
DisasterM3 Bench (list only)	2,679	2,199	856	Photos are not on our computer. The paper says 5,024 color and 976 SAR.
BRIGHT	3,395 here (4,246 in the full release)	0	4,246	Ukraine, Myanmar and Mexico pre photos are missing

Tags, black photos, missing files, event names

original p. 3

"Tags are right; file names are not."

- The DisasterM3 tags and the pixel test agree on every post photo.
- But 2,584 of the 3,537 SAR posts (73%) have **no "sar" in the file name**.
- Example: `la_palma_volcano_00000002_post_disaster.png`. This file is SAR, but its name does not say so.
- So if you remove SAR by name, these SAR photos stay in your color data.
- The old script `dm3_bright_audit.py` finds SAR by name, so it makes this mistake.

"Eight pre images are over 90% black"

- 8 pre photos are more than 90% black: 1 from `rwanda_volcano`, 4 from `spain_volcano` and 3 from `ukraine_conflict`.
- A black pixel has red = green = blue = 0. So if you test the whole photo, it looks like SAR.
- But the few real pixels in these photos have color. This is why our rule checks only the pixels that have data.
- None of these 8 photos is in the optical pool, so there is no problem.

"2,066 unreferenced files": 2,066 files are not in any list. Almost all of them come from BRIGHT events: `spain_volcano` 478, `la_palma_volcano` 367, Morocco under 3 different spellings 617, and `libya_flood` 126.

"36 event names": DisasterM3 has 36 events, the same number as its paper.

- **19** come from xBD (another dataset).
- **14** come from BRIGHT. Two of them have new names: "Spain volcano" is La Palma, and "Rwanda volcano" is Goma.
- **3** come from somewhere else: Kalehe, Shovi and Hurricane Ian.

The paper says 10 events are new, but only these 3 are really new. Also, 5 events are **only in Bench**: Guatemala volcano, Joplin tornado, Mexico earthquake, Hurricane Ian and Noto earthquake.

Resolution and geometry: photo detail and size

original p. 3–4

"BRIGHT's own GeoTIFF tags put its tiles at 0.2–0.6 m per pixel..." The BRIGHT files say 0.2 to 0.6 m per pixel. DisasterM3 has 2,828 copies of BRIGHT tiles, and they are still **0.30 m**, not 0.8 m as its paper says.

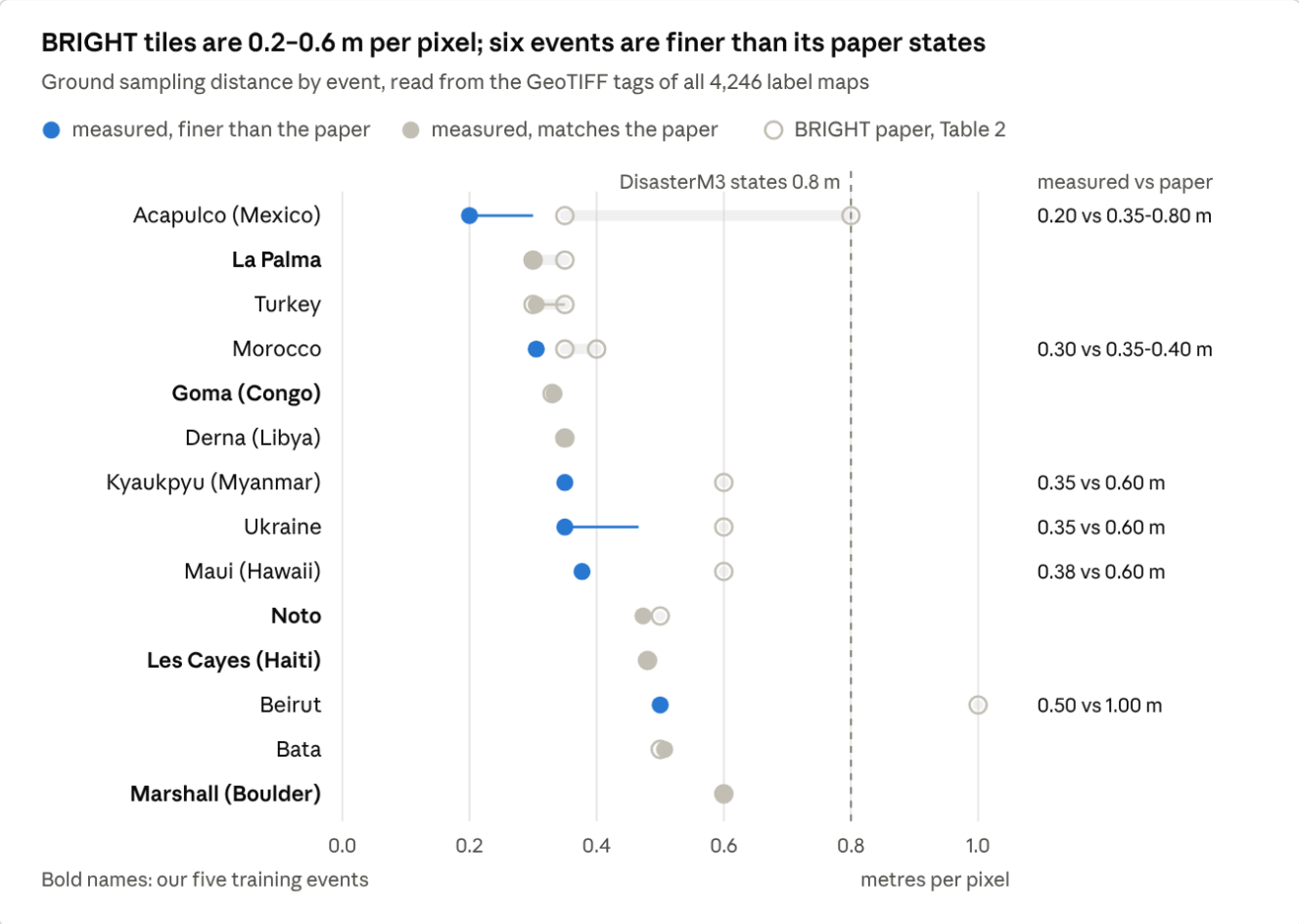


Chart from the original PDF (p. 4). Source: dm3_bright_optical_eda.py, the GeoTIFF tags of 4,246 BRIGHT label maps, BRIGHT paper Table 2, DisasterM3 paper §3.

> **How to read this chart.** It shows the resolution of each event. "Finer" means more detail.
>
> - 🔵 **Blue dot**: our measurement, with more detail than the paper says.
> - ⚪ **Grey dot**: our measurement, the same as the paper.
> - ○ **Empty circle**: what the BRIGHT paper says (its Table 2).
> - **Dashed line at 0.8**: what the DisasterM3 paper says.
> - **Text on the right**: our number vs the paper's number, for each event that does not match.
> - **Bold names**: our 5 training events (La Palma, Goma, Noto, Les Cayes, Marshall).

"Beirut is the widest gap": Beirut has the biggest difference: 0.50 m in the file, 1.0 m in the paper.

"Tile size." Every color photo in both datasets is **1024 × 1024 pixels**: 17,451 in DisasterM3 and 3,395 in BRIGHT.

"DisasterM3's 0.8 m."

- The DisasterM3 paper says it changed every photo to 0.8 m. But its 2,828 copies of BRIGHT tiles are 0.30 m.
- DisasterM3 photos have no location or scale information, so we **cannot measure** the other photos.

"Sharpness." We measure how many small, sharp details a photo has. A blurry photo scores near 0.

Photos	Sharpness (middle value)
Pool, post photos	0.073
Pool, pre photos	0.060
BRIGHT, 11 events	0.067
Our 5 events	**0.189** (La Palma alone: 0.223)

- Our 5 events are much sharper than the rest.
- The Turkey (0.011) and Morocco (0.009) post photos are the **most blurry** in the pool. But their pre photos are BRIGHT tiles, which are 3 to 9 times sharper.

"Object scale." How many pixels does one building cover? (middle value)

- In the pool: 1,030 pixels. In all of BRIGHT: 945 pixels.
- In our 5 events: **471 pixels**. La Palma is 635 pixels (about 57 m^2), and Goma is 422 pixels (about 47 m^2).

> **Meaning:** In our 5 events, buildings look **small** in the photo.

Brightness, color and how different the photos look

original p. 5

The original title is **"Radiometry and domain shift"**.

"Our five BRIGHT events are the furthest from the images..."

- Our 5 events look the **most different** from the photos the model learns from and is tested on.
- They are darker, with less color, and sharper.
- The difference score is 0.11 bits. (0 means the same, 1 means totally different.)

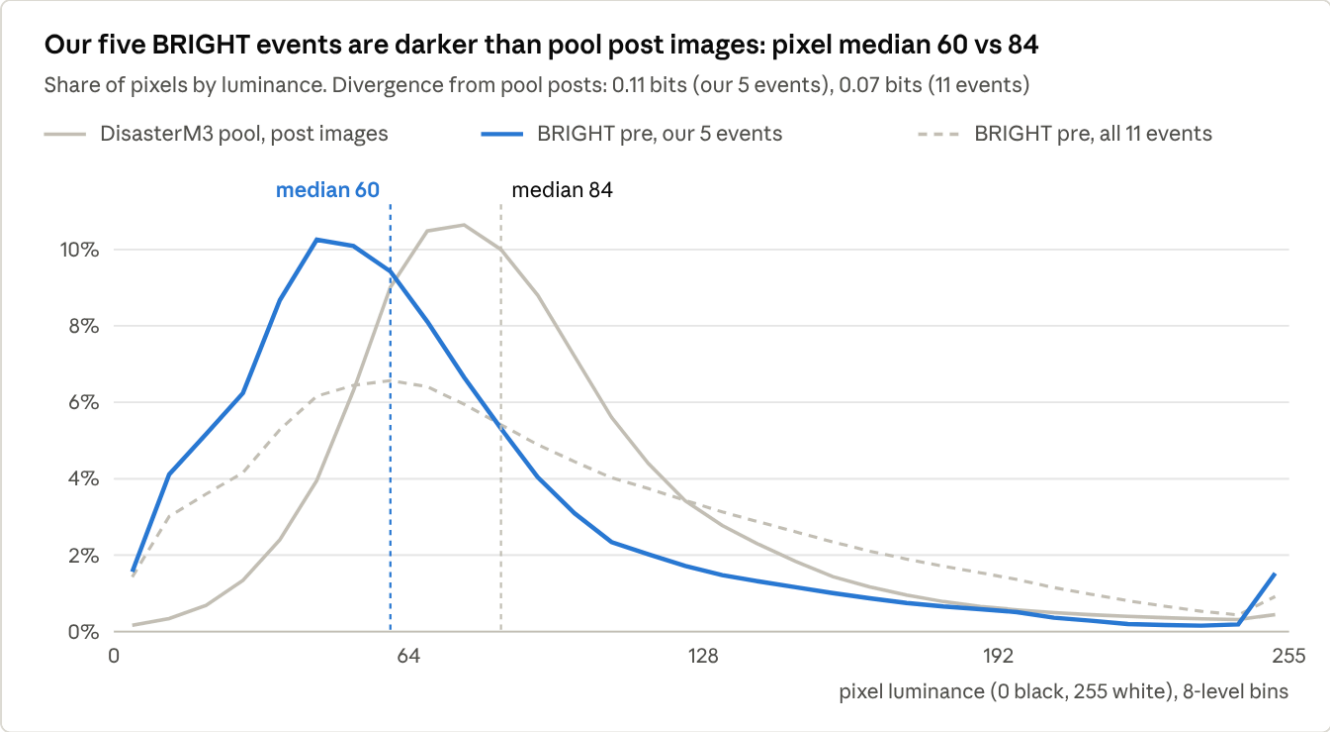


Chart from the original PDF (p. 5). Source: dm3_bright_optical_eda.py, brightness of non-black pixels in 6,443 pool post photos, 3,395 BRIGHT pre tiles and the 1,285 tiles of our 5 events.

How to read this chart.

- Left to right: pixel brightness (0 = black, 255 = white). Up: what % of pixels have that brightness.
- **Grey line** = pool post photos. Their middle brightness is 84.
- **Blue line** = our 5 events. Their middle brightness is 60. The blue line is more to the left, so these photos are **darker**.
- **Dashed line** = all 11 BRIGHT events.

"Most of the gap is La Palma's dark volcanic ground"

- Most of this difference comes from La Palma, because volcano ground is dark (brightness 60).
- But DisasterM3 events already go from 61 (Florence) to 130 (Maui). So dark photos are not new to the model.

"Pre vs post inside DisasterM3."

- If we put all pre photos together and all post photos together, they look almost the same (0.002 bits).
- But one pair at a time, they are different: the middle difference is 0.057 bits.
- In 18% of pairs, the difference is more than 0.2 bits. The reasons are floods (water), burn marks, a different satellite, or a different season.

"Colour." Color strength: pool post photos 0.30, our 5 events 0.21. So our photos have less color. Middle photo brightness: 89 vs 66.

Note: "60 vs 84" and "66 vs 89" are two different ways to measure.
60 vs 84: we put all pixels together, then take the middle value.
66 vs 89: we find each photo's brightness first, then take the middle value.

"No-data." (black empty areas) 10.3% of pool photos and 15.9% of BRIGHT tiles are **more than 10% black**. The worst BRIGHT events are Bata (28% of tiles), Goma (28%), Marshall (27%) and Derna (26%).

"Cloud, haze, smoke." 139 pool **post** photos are more than 20% covered by cloud, haze or smoke. Only 15 **pre** photos are.

> **Meaning:** Bad weather on the disaster day hurts the **post** photos most, and the model reads the post photos.

Do the pre and post photos line up? original p. 6

The original title is **"Pre/post registration"**.

"Half of DisasterM3's measurable optical pairs are misaligned by more than 2 px..."

- In half of the pairs we could measure, the photos are moved by **more than 2 pixels**. The middle value is 2.1 pixels.
- The paper says "strictly aligned". That is **not true**.
- How far the photos move depends on the event.

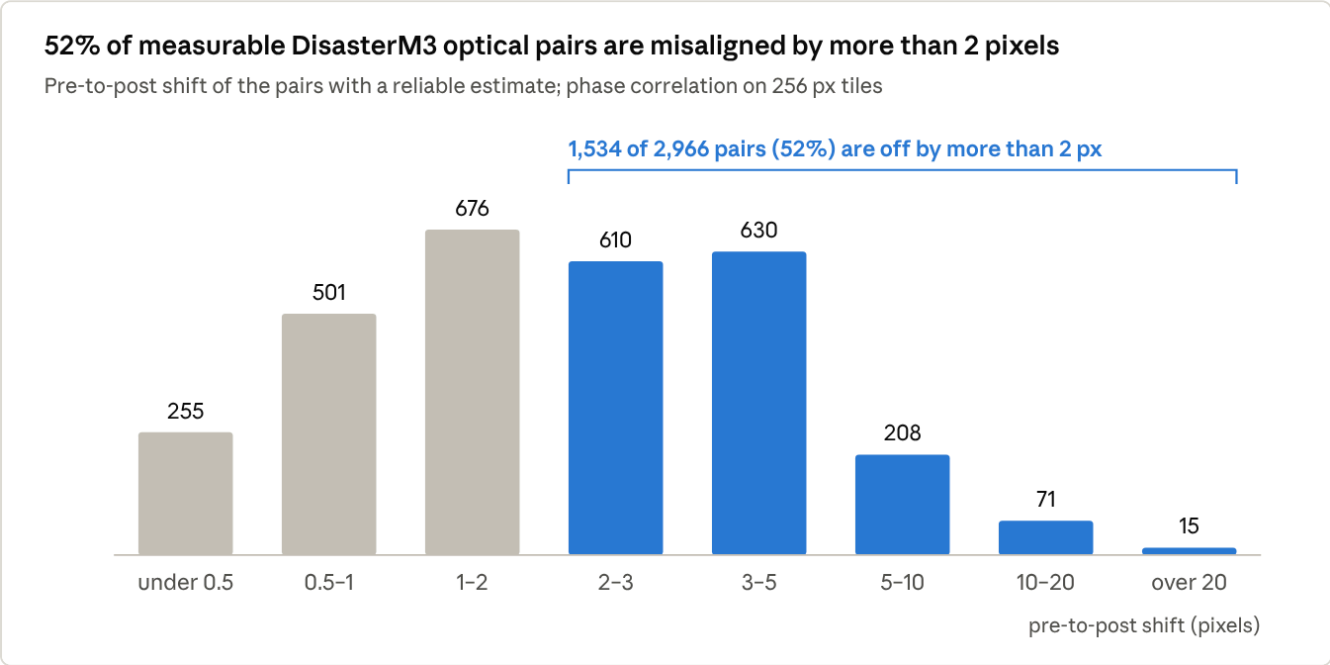


Chart from the original PDF (p. 6). Source: dm3_bright_optical_eda.py, 2,966 of DisasterM3's 7,159 color pairs with a shift we can trust (at least 4 pieces agree within 1.5 px).

> **How to read this chart.** Each bar shows how many pairs moved by that many pixels. The **blue bars** are the pairs that moved more than 2 pixels: together **1,534 of 2,966 pairs = 52%**.
>
> We used "phase correlation". This is a math method that finds how far one photo moved. It works on pieces of 256 pixels.

"Portugal wildfire pairs are nearly aligned..." Portugal wildfire photos are almost correct: they moved 0.8 px. Hurricane Harvey moved 7.1 px, Turkey scene 7 moved 10.5 px, and Morocco moved 18.7 px.

"Coverage."

- We could measure only **2,966 of 7,159 pairs (41%)**.
- For the other pairs, the photos could not be matched because of real changes (broken buildings), cloud or black areas.
- 47 pairs have no usable piece at all.

"Worst cases." `turkey_earthquake1` pairs 507, 516 and 522 are moved by **40–56 px**. `hurricane_florence` pairs 107, 150, 427 and 476 are moved by 26–31 px.

"Who is affected."

- Our champion model looks **only at post photos**, so this problem does **not** hurt it.
- A model that uses both pre and post photos (Siamese or change detection) **must line the photos up first**.

Labels: how many of each kind

original p. 6–7

The original title is **"Labels and class balance"**.

"Destroyed buildings are scarce where the model is scored..." In the test set, destroyed buildings are **very few**:

- Only **3.6%** of building pixels are destroyed.
- **66.5%** of test photos have no destroyed pixel at all.

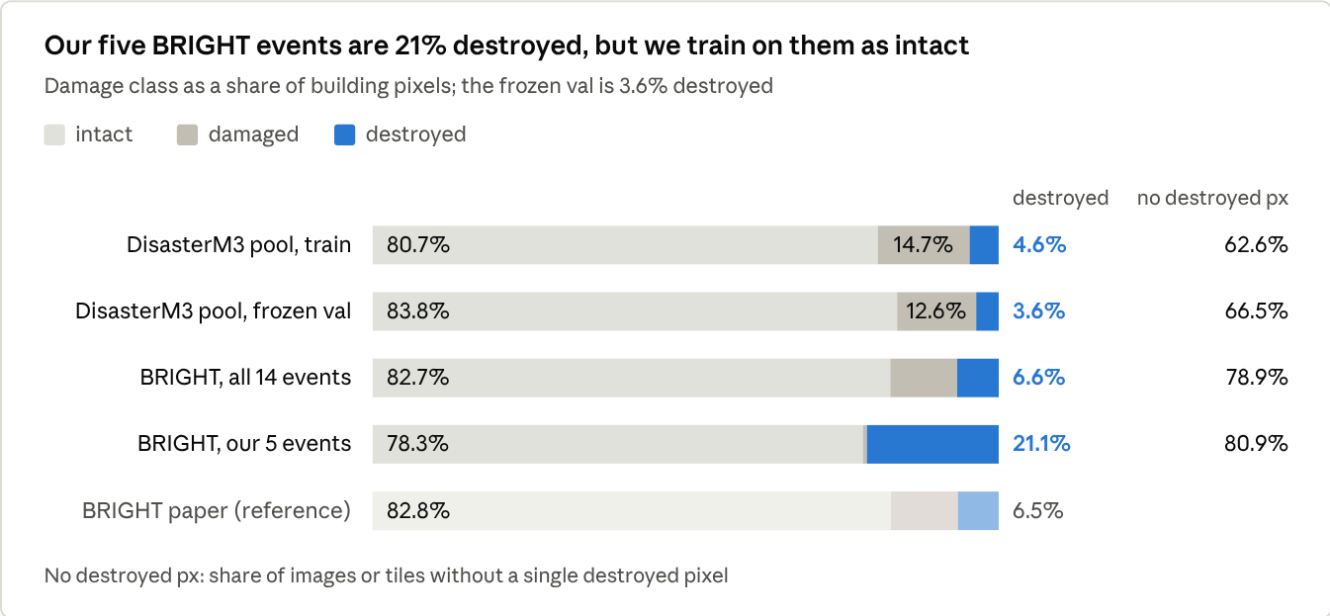


Chart from the original PDF (p. 7). Source: dm3_bright_optical_eda.py, the 4-class maps of the 6,443 pool photos and the 4,246 BRIGHT label maps, BRIGHT paper Fig. 5d.

Group	Intact	Damaged	Destroyed	Photos with no destroyed pixel
Pool, learning set	80.7%	14.7%	4.6%	62.6%
Pool, test set	83.8%	12.6%	3.6%	66.5%
BRIGHT, all 14 events	82.7%	about 10.7%	6.6%	78.9%
BRIGHT, our 5 events	78.3%	almost 0	**21.1%**	80.9%
BRIGHT paper	82.8%	about 10.7%	6.5%	–

> **Meaning:**
>
> - Our 5 events have many destroyed buildings (21%).
> - But 80.9% of their tiles have no destroyed pixel. So the destroyed buildings sit in **a few tiles only**.
> - And we only have the **pre** photos, when the buildings were still fine. So we train on them as **intact**.

"BRIGHT's label maps match its paper..." BRIGHT's labels match its paper: 14.1% building pixels, and 82.7 / 10.7 / 6.6%. So the **labels are fine**. The problem is the **photos**: we only see the "before" view.

"The merge is clean." Our notebooks join 3 masks (intact, damaged, destroyed) into one map. No pixel is in two masks at the same time, and every mask has only 0 or 255. So the order of joining **does not matter**.

"Spread by event." The destroyed % changes a lot between events, from 0.07% (Turkey scene 0) to 63% (Maui).

"Object counts." The pool has 249,623 buildings (middle size 1,030 px). BRIGHT's maps have 211,318, but its paper says 384,596. Why? Buildings that touch each other are **counted as one**. Example: in Turkey we count 66,055, but the paper says 135,033.

"Empty tiles." 839 BRIGHT tiles (19.8%) have **no building at all**. 315 of them are in our 5 events (24.5% of their tiles). No pool photo is empty.

"Roads." Pool road pixels are 92.0% fine, 6.6% under flood water and 1.4% covered by broken material. Our model **does not use roads**, so this is only information.

Which events are in both datasets? original p. 7–9

The original title is **"Events and overlap"**.

"All 14 BRIGHT events are inside DisasterM3, but only six reach its optical pool..."

- All 14 BRIGHT events are also in DisasterM3. But only **6** of them are in the optical pool.
- 3 of those 6 (Turkey, Morocco, Derna) use **exact copies** of BRIGHT pre photos.

What each column means: (1) the event name in BRIGHT; (2) the name used in DisasterM3, and some are spelled wrong; (3) how many pairs are in the pool, as learning / test; (4) how many SAR pairs there are; (5) how many BRIGHT tiles have an exact copy in DisasterM3.

BRIGHT event	Name in DisasterM3	Pool pairs (learn / test)	SAR pairs	Exact copies of BRIGHT tiles
Turkey	`turkey_earthquake0-7`	557 / **64**	580	583 of 1,114; 579 are pool pre photos
Morocco	`morocco_earthquake`, `morroco_earthquake1` (spelling mistake)	58 / **3**	453	428 of 567; 61 are pool pre photos
Derna	`libya_flood`	43 / **4**	51	52 of 124; 47 are pool pre photos
Beirut	`beriut_explosion` (spelling mistake)	23 / 4	27	0 of 133
Bata	`bata_explosion`	23 / 3	26	0 of 107
Maui	`hawaii_wildfire`	14 / 1	9	0 of 65
La Palma ours	`la_palma_volcano`, `spain_volcano`	0 / 0	1,652	921 of 933, but only inside SAR pairs
Goma ours	`rwanda_volcano`	0 / 0	93	93 of 123
Marshall ours	`marshall_wildfire`	0 / 0	17	17 of 77
Les Cayes ours	`haiti_earthquake`	0 / 0	8	8 of 73
Noto ours	`noto_earthquake` (Bench only)	0 / 0	0	0 of 79
Ukraine	`ukraine_conflict`	0 / 0	413	Could not check: no pre photos on our computer
Acapulco	`mexico_hurricane`	0 / 0	112	Could not check: no pre photos on our computer
Kyaukpyu	`myanmar_hurricane`	0 / 0	96	Could not check: no pre photos on our computer

"Our five events cannot leak into train or val." Our 5 events have **no pair in the pool**. Their copies in DisasterM3 are only SAR pairs and Bench masks. So they **cannot cause a leak**.

"Turkey, Morocco and Libya would leak."

- 66 test pairs use a BRIGHT tile from Turkey, Morocco or Libya as their pre photo.
- If we put these tiles in the learning set, the model will see test photos. That is a leak.
- So **keep these tiles out of learning**.

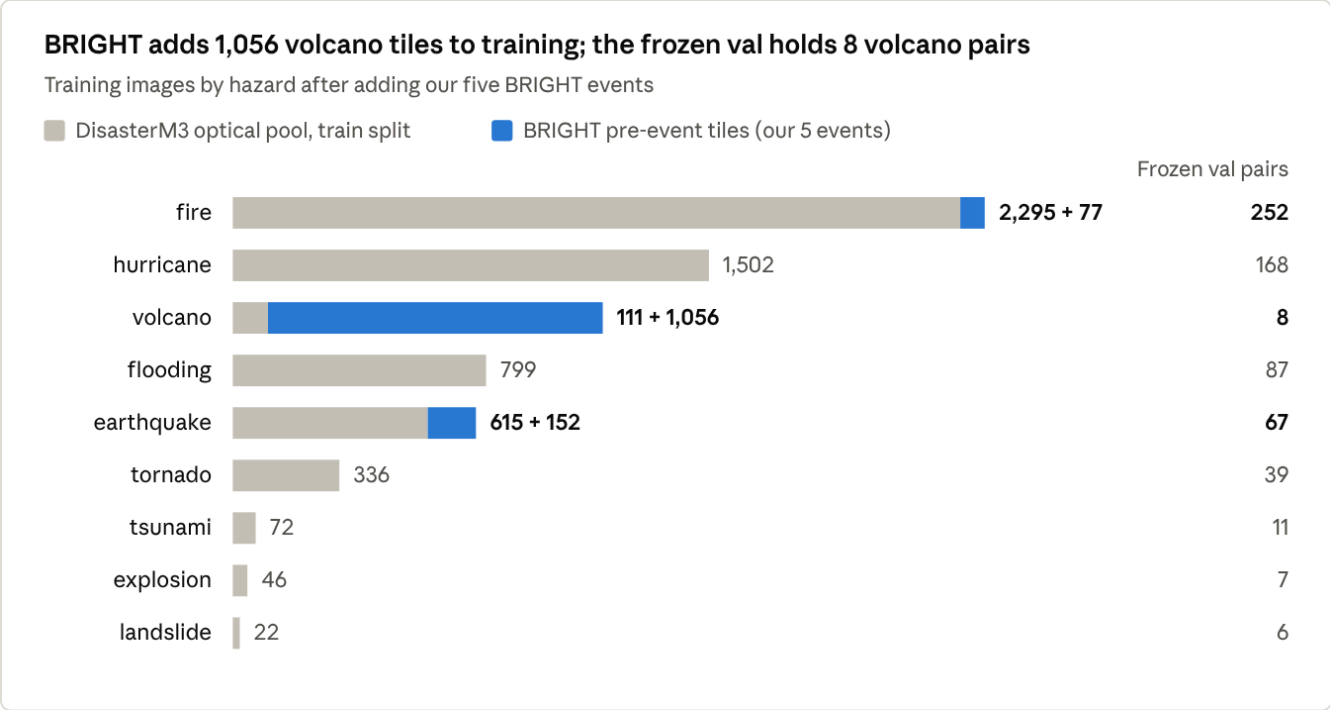


Chart from the original PDF (p. 9). Source: dm3_bright_optical_eda.py, the pool learning set (5,798 pairs), the frozen test set (645), and the BRIGHT pre tiles of La Palma, Goma, Les Cayes, Marshall and Noto (1,285).

> **How to read this chart.** It shows the learning photos for each disaster type. **Grey** = DisasterM3. **Blue** = our 5 BRIGHT events. The right column is the number of test pairs for each type.
>
> Fire gets +77 (Marshall). Volcano gets **+1,056** (La Palma + Goma). Earthquake gets +152 (Les Cayes + Noto).

"Our five events add data almost only where the val measures least..." 1,056 of our 1,285 tiles are volcano photos. But the test set has only 8 volcano pairs (out of 645).

> **Meaning:** We add data where the test **almost never looks**. So the score will **not change much**.

"Duplicates inside DisasterM3." (the same photos twice)

- `la_palma_volcano` and `spain_volcano` hold **the same La Palma pairs two times**. That is 1,456 identical photos, all in SAR pairs.
- `socal_fire` and `woolsey_fire` share **264 identical pre photos** inside the optical pool. The next section shows what this does to the test set.

Checking the learning / test split

original p. 9–10

The original title is **"Frozen-split audit"**.

"The 645-pair validation set mirrors the train split..." The test set looks like the learning set: similar events and similar labels. But 40 test pairs (6.2%) show **places that are also in the learning set**, and they have 8.6% of the test's destroyed pixels.

Measure	Learning	Test	Meaning
Pairs	5,798	645	
From xBD events	87.2%	86.8%	almost the same
Building pixels, out of all pixels	7.21%	7.45%	almost the same
Damaged pixels, out of building pixels	14.67%	12.63%	a little less in test
Destroyed pixels, out of building pixels	4.63%	3.59%	a little less in test
Photos with no destroyed pixel	62.6%	66.5%	

> **Meaning:** The split is mostly fair. The only problem is the 40 leaked pairs.

"How the leak happens."

- `socal_fire` and `woolsey_fire` have the same 264 pre photos, under two names.
- The random split put 40 of them in the test set, and their **"twin"** in the learning set.
- The twins are very alike:
  - The pre photo is **exactly the same**.
  - The post photos are **88% similar** (r = 0.88).
  - The building masks **overlap 72%** (IoU 0.72).
  - On 97% of the shared building pixels, the damage label is **the same**.

"Why a name check misses it." The file names are all different. Only the **pixels** are the same. So checking names does not find the leak.

"What it does to the gate."

- The test set has 1,808,111 destroyed pixels, and 155,812 of them are in these 40 pairs.
- So the 645 score is **a little too good**. The model gets points just for **remembering** the twins.
- **Fix:** also report the score on the **605 clean pairs**. The list of the 40 leaked pairs is in `val_train_leaks.csv`.

"Every val event also appears in train..."

- Every test event is also in the learning set. This is normal for a random split.
- Tiles that sit **next to each other on the ground** could also be split between learning and test. That is another kind of leak.
- But DisasterM3 photos have no location, so we **cannot check this**.

Problems in the data original p. 10–11

The original title is **"Data-quality issues"**. We found 11 problems. **Only the first one** makes the test score wrong. The others change counts, filters, or how we prepare photos.

#	Problem	How many	Effect on us
1	The same place in learning and test (socal = woolsey)	40 test pairs, 8.6% of destroyed pixels	The 645 Destroyed IoU looks **too good**

#	Problem	How many	Effect on us
2	Pre and post photos do not line up	52% over 2 px, 10% over 5 px, worst 55.6 px	No effect on the champion. A pre+post model must fix it.
3	More than 10% black area	10.3% of pool photos, 15.9% of BRIGHT tiles	Wasted crops. Brightness fixing goes wrong.
4	Blurry, stretched post photos	Turkey posts have 3 times less detail than their pre photos, Morocco 9 times less	Blurry posts next to sharp 0.30 m pre photos
5	Cloud, haze or smoke over 20% of the tile	139 post photos, 15 pre photos	It **hides the damage** in the photos the model reads
6	SAR files without "sar" in the name	2,584 of 3,537	Name filters let SAR into color data
7	La Palma saved under two names	1,456 identical photos	La Palma is counted twice
8	Wrong or mixed spelling of event names	`beriut`, `morroco`; Morocco under 3 names, Turkey under 6	Grouping events by name goes wrong without a name map
9	Files not in any list	2,066	Loaders that read the list never see them
10	BRIGHT tiles with no building	839 of 4,246; 315 of our 1,285	Learning photos with only background
11	The papers give the wrong resolution	BRIGHT: 6 of 14 events. DisasterM3 says 0.8 m, but its BRIGHT copies are 0.30 m.	Plans based on the papers will be wrong

"Eight pre images are over 90% black no-data; none reaches the optical pool." 8 pre photos are almost all black, but none of them is in our pool. No problem.

What to do

original p. 11–12

The original title is **"Recommendations"**.

> **First:** fix the leak. Adding BRIGHT is safe, but **do not expect** it to raise the Destroyed score.

1. **"Report the gate twice."**
   - Keep the score on all 645 pairs, so you can compare it with the old runs R1–R6.
   - Also report the Destroyed IoU on the **605 clean pairs** (the pairs not in `val_train_leaks.csv`).
   - If you make a new split, group the pairs by the "fingerprint" (hash) of the pre photo. Then twins always stay on the same side.
2. **"Keep BRIGHT Turkey, Morocco and Libya out of training."** 66 test pairs use these tiles as pre photos. Our 5 events are fine: none of their tiles is in the pool.

3. **"Set expectations for the five events."**
   - They add pre-disaster photos with buildings labeled "intact".
   - 82% of them are volcano photos, and the test has only 8 volcano pairs.
   - Their labels say 21% destroyed. But there is no color post photo of these events on our computer.
   - The only one anywhere is Noto: 68 photos in DisasterM3 Bench.
4. **"Match appearance before mixing."** Make the photos look alike before you mix them.
   - The problem: our 5-event photos are darker (66 vs 89) and sharper, and their buildings are smaller (471 vs 1,030 px).
   - Fix the brightness of each photo by itself (per-image normalization).
   - Add blur and color changes (augmentation).
   - Think about making these photos **1.5 times bigger**. Why? The buildings are about 2.2 times smaller in area. If the width is 1.5 times bigger and the height is 1.5 times bigger, the area grows 1.5 × 1.5 = 2.25 times. Then the building sizes match.
5. **"Filter SAR by tag or pixels, never by file name."** Stop using the name-based SAR check in `dm3_bright_audit.py` and `dm3_vs_bright_compare.py`.
6. **"Skip mostly-black crops."** 10–16% of tiles are more than 10% black. When you cut a random piece (crop) from a photo, **throw it away if more than half of it is black**.
7. **"Align pairs before any pre+post model."** The champion uses only post photos, so it is fine. A Siamese model (pre + post) would see photos moved by about 2 px, and by more than 5 px in 1 of every 10 pairs.
8. **"Take resolution from GeoTIFF tags, not the papers."** BRIGHT's Table 2 is wrong for 6 events. DisasterM3's BRIGHT copies are 0.30 m, not 0.8 m.

How to make the numbers again

original p. 12–13

The original title is **"Reproduce and definitions"**. Run these 2 commands to make **every number again**. It takes about 35 minutes on a laptop with 8 GB of memory.

```
cd Essential_task_delegation/BRIGHT_DisasterM3_Integration
python dm3_bright_optical_eda.py   # the main check; results go to data/dm3_bright_optical_eda/
python bright_dm3_dedup.py         # checks every BRIGHT tile against the pool
```

Output files: what each file has.

File	One row =
`summary.json`	**every number** in this report
`dm3_images.csv`	one DisasterM3 photo: pre or post, event, tag, SAR or color, brightness, fingerprint (md5), split
`bright_pre_tiles.csv`	one BRIGHT pre tile: resolution, brightness, fingerprint
`bright_targets.csv`	one BRIGHT label map: pixels of each label, buildings, resolution
`dm3_pool_labels.csv`	one pool photo: pixels of each label, buildings, road pixels

File	One row =
`dm3_optical_pair_registration.csv`	one pair: how far it moved, how many pieces agreed, whether we can trust it
`dm3_pool_pairs_radiometry.csv`	one pool pair: pre and post brightness, color, difference
`dm3_events.csv`	one event name: disaster type, where it came from, counts
`val_train_leaks.csv`	one leaked test pair: its twin in the learning set, and how similar they are

Definitions

- **"High-frequency share"**: how much small, sharp detail is in the middle 512 pixels of a photo. Blurry or stretched photos score **near 0**.
- **"JS divergence"**: a number for how different two brightness patterns are. **0 = the same, 1 = totally different.**
- **"Reliable shift"**: we cut the photos into pieces of 256 px and measure how far each piece moved. We trust the result only if **at least 4 pieces agree** (within 1.5 px), and each one gives a clear signal (a peak of 0.1 or more).
- **"Building components"**: a group of building pixels that touch each other. Buildings that touch are counted as **one**, so in busy areas we count **too few**.

Part 2 · Mistakes in the older PDF

original p. 14–17

The original title is **"Earlier PDF: corrections"**.

- There was an older 4-page PDF (*EO Data Audit & Exploratory Data Analysis: DisasterM3 vs. BRIGHT*).
- It copied the score tables from the papers **correctly**.
- But **most of its data numbers are wrong**, or they do not match the files on our computer.

The old PDF said	Right or wrong?	The truth
"4,246 BRIGHT pre-event EO tiles"	Wrong for the release	Only 3,395 color pre tiles. The other 851 (Ukraine, Myanmar, Mexico) are Google Earth photos. You must download them yourself with `aoi_reference_kmz.zip` and BRIGHT's guide.
BRIGHT "0.3–1.0 m"	Wrong	0.2–0.6 m
"22,214 DisasterM3 images (0.8 m)"	Half right	The number is from the paper. On our computer: 17,451 color photos (the paper says 17,190). The BRIGHT copies are 0.30 m, not 0.8 m.
"11,107 color pairs"	Wrong	Even the old PDF's own math gives 17,440 / 2 = 8,720. The lists have 7,159 (Instruct) + 2,199 (Bench) = 9,358.
"384,596 checked buildings"	Not checked	This number is from the paper. The paper's own rows add up to 358,340. The label maps have 211,318 (touching buildings count as one).
BRIGHT "5-event color post subset"	Not released	It was used only in one experiment in the BRIGHT paper (section 4.4). All released post photos are SAR.
DisasterM3 "all photos changed to 0.80 m"	Wrong	2,828 photos are exact BRIGHT copies at 0.30 m
DisasterM3 "has location, 0.8 m grid"	Wrong	DisasterM3 photos have **no** location information
"xBD 15 + BRIGHT 11 + 10 new events (with Nepal flooding)"	Wrong	19 xBD + 14 BRIGHT + 3 new (Kalehe, Shovi, Hurricane Ian). Nepal flooding is an xBD event.
DisasterM3 "instance masks"	Not exact	One black-and-white mask for each label. You can find single buildings only by grouping touching pixels.
Resolution column	Wrong for 7 of 14 events	Beirut 0.50 (not 1.00), Ukraine 0.35–0.47 (not 0.60), Myanmar 0.35 (not 0.60), Maui 0.38 (not 0.60), Morocco 0.31 (not 0.35–0.40), Acapulco 0.20–0.30 (not 0.35–0.80), Noto 0.47 (not 0.50)
Color tiles: Ukraine 513, Myanmar 126, Acapulco 212	Not released	0 in `pre-event.zip`. DisasterM3 has 410, 96 and 112 of them, with BRIGHT's labels.
"Reg. RMSE" column	SAR information	It is about lining up color photos with SAR (BRIGHT Table C1). For color pre/post photos: 52% of pairs are moved more than 2 px.

The old PDF said	Right or wrong?	The truth
"5 events (488 tiles)" with color post photos	Wrong sum, not released	133 + 107 + 65 + 124 + 79 = **508**. No color post photo was released.
BRIGHT tiles "made smaller to 0.80 m", "2.28–2.67 times less detail"	Wrong idea	DisasterM3 keeps them exactly the same, at 0.30 m (La Palma, Goma, Les Cayes, Marshall, Turkey, Morocco, Derna)
BRIGHT 85.9 / 14.1%, 6.1 : 1	Right	14.14% buildings, 6.07 : 1
BRIGHT 82.8 / 10.7 / 6.5%	Right	82.7 / 10.7 / 6.6%
DisasterM3 92.85 / 1.81 / 5.34%, "damaged is 5.9 times less"	Masks disagree	In the pool, building pixels are 81.0 / 14.5 / 4.5%. Out of all pixels: 5.86 / 1.05 / 0.33%, the same as our Integration Plan. DisasterM3 actually has **more** damaged than BRIGHT.
Roads 79.44 / 12.94 / 7.62%	From the paper	In our pool: 92.0 / 6.6 / 1.4%
27,074 vs 26,988 "because of 86 multi-hazard tiles"	No proof	The paper gives no reason. 86 is just the difference between the two numbers.
Photo counts by disaster type (6,750 fire …)	Right (paper Fig. 2)	But these mix color + SAR, and Instruct + Bench. Our color pool only: fire 2,547, hurricane 1,670, flooding 886, earthquake 682, tornado 375, volcano 119, tsunami 83, explosion 53, landslide 28, conflict 0.
BRIGHT Table 7 gains (F1 68.71 → 73.59 …)	Right, but SAR	Table 7 uses SAR post photos. So these are color + SAR results, not color only.
BRIGHT Table 8, DisasterM3 Tables 2, 3, 9	Right	All numbers match the papers. Only one label is wrong: "Intact IoU" is put under "Counting".
+3.80 mIoU, +8.32 Destroyed IoU, +5.7 mIoU, +7.9 Building IoU, 5.4 times fewer parameters	Right	Checked with BRIGHT Table 8 and DisasterM3 Table 3
Pages 2–4	Text is cut	Sections 3, 5A and 7 stop in the middle of a sentence

"Missing from the PDF": things the old PDF did not have

- The test leak: 40 pairs whose twins are in the learning set, with 8.6% of the destroyed pixels.
- All 14 BRIGHT events are inside DisasterM3, and the Turkey, Morocco and Libya tiles are pre photos of 66 test pairs.
- 73% of SAR files have no "sar" in the name.
- The pre/post photos do not line up, and the BRIGHT photos look different.

"Suggested fixes": how to fix the old PDF

1. Replace sections 1–5 with the numbers we measured. Put "what the paper says" and "what we measured" in **two separate columns**.
2. Keep sections 6–7, but fix the wrong column label.
3. Remove the SAR content (Table 7, Table C1), or label it "color + SAR".

4. Next to every number from a paper, write its table or figure number.
5. Export the PDF again, so no section is cut.

The whole report in 5 lines

1. **There is a leak:** 40 test pairs are copies of learning pairs. Report the score two times: on 645 pairs and on 605 clean pairs.
2. **Do not use** the BRIGHT Turkey, Morocco and Libya tiles for learning.
3. **Our 5 BRIGHT events are safe**, but they will **not** raise the Destroyed score much.
4. **Find SAR by tags or pixels**, never by file name.
5. **Read the resolution from the file tags**, not from the papers.